

M002 Week 7 — English Worksheet (Beginner)

Topic: Pistons and Mazes · **Course:** Maze Madness · **Level:** Beginner · **Time:** about 30 minutes

This week the maze uses a **piston**. When your agent flips a lever, the piston opens a path across a gap. The agent crosses while the path is open.

1 · Predict 🧠

Read the steps. Before you imagine the agent doing it, circle or write your answer.

```
move forward
interact ahead
move forward
```

The agent flips the lever and the piston opens the path. Can the agent cross now? Circle one:
yes · no


```
if gap ahead:
    interact ahead
otherwise:
    move forward
```

The agent is on solid ground with no gap. What does it do? Circle one: interact · move forward

2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — The agent should power the piston **first**, then cross.

```
# clean
interact ahead
move forward
move forward
```

```
# buggy
move forward
move forward
interact ahead
```

What is wrong? What happens at the gap?

Pair B — The agent should **interact** only when there is a **gap** ahead.

```
# clean
if gap ahead:
    interact ahead
otherwise:
    move forward
```

```
# buggy
keep doing forever:
    interact ahead
    move forward
```

What is wrong? How often does the buggy agent interact?

3 · Fill the Gap

The agent should flip the lever **only** when it sees a gap. One word is missing. Fill it in using the word bank.

```
if ____ ahead:
    interact ahead
otherwise:
    move forward
```

Word bank: gap · grass · friend

Write the missing word:

4 · Tell Me What You Built 📷

Now switch to your homework world. The maze has a **gap** and a **piston** that opens it. Walk the agent to the lever, flip it, cross while the path is open, and reach the goal. When you finish, come back here.

Send a photo or video of the agent reaching the end, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

I powered the piston by ...

Once the path opened, the agent ...

The hardest part was ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while the agent runs the maze. Talk like you are teaching a friend. Try to use these words: **piston**, **lever**, **gap**, **bridge**.

1. Show the gap and the piston.
2. Run your code and show the agent crossing.
3. Say which line of your code opens the path.

Write what you will say in your video. You can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.